LIAM COULTER

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WORK EXPERIENCE

RTX BBN Technologies St. Louis Park, Minnesota *August 2023 - Present*
Sr. Engineer (Research), Spectrum Dominance group

- Participated in multiple field demonstrations of experimental technology to end customer including preparation, test planning, and resolving unexpected issues on-site. Test locations: government and RTX sites
- Prepare and participate in periodic and standalone project review presentations to end customer, including kickoffs, quarterly progress updates, and final reports
- Contribute to project proposals and presentations for new & existing external customers, to win new work and maintain strong relationships
- Member of Spectrum Dominance group focused on RF, signal processing, and signal intelligence, within Non-kinetic Capabilities and Technologies business unit. Concurrently pursuing PhD

RTX BBN Technologies St. Louis Park, Minnesota *May 2020 - August 2023*
Staff Scientist I, Spectrum Dominance group

- Drive program execution for a wide range of government customers, as well as Raytheon internal research & development projects
- Develop & implement algorithms for signal and image processing across a wide range of technical areas including passive & remote sensing, distributed systems, Synthetic Aperture Radar (SAR), geolocation & target tracking, and atmospheric signal propagation modeling

NASA Goddard Space Flight Center Greenbelt, Maryland *August 2019 - May 2020*
Graduate Student Researcher, Supervisor: Dr. Joel McCorkel

- Developed experimental data processing chain and Python implementation of radio occultation using geostationary satellite, with real NASA data from GOES-16 spacecraft
- Presented results at American Meteorological Society Fifth Conference on Atmospheric Biogeosciences, June 2021 (abstract published), and received "Outstanding Oral Presentation" award

Autonomy Technology Research Center Dayton, Ohio *May 2019 - August 2019*
Graduate Student Research Intern, Supervisor: Dr. Chad Waddington

- Implemented and compared algorithms for passive radar emitter localization (Python & MATLAB)
- Utilized high-performance computing clusters for simulation experiments, presented results to senior Air Force Research Laboratory scientists

Graco, Inc. Minneapolis, Minnesota *May 2018 - March 2019*
Big Data Graduate Intern

- Designed custom implementation of Kalman filter-based prediction algorithm for inventory forecasting (Python & R) which out-performed existing business methods; presented results to Graco senior leadership
- Identified data analytics software for Business Intelligence team, which Graco purchased, and implemented classification algorithms on company data to predict future part failures & maintenance needs

EDUCATION

University of Minnesota Minneapolis, Minnesota *September 2020 - Present (Expected Summer 2026)*
PhD Candidate, Electrical and Computer Engineering. Advisor: Dr. Jarvis Haupt GPA 3.88

- Developing novel approaches to imaging without a line-of-sight, and investigating fundamental bounds for recovering data from light which has undergone diffuse reflection
- Classwork: estimation & detection theory, compressed sensing, stochastic processes, image processing, parallel computing, and data compression

- PhD candidate, planned graduation 2026. Concurrently working (see above)

Colorado State University Fort Collins, Colorado *August 2017 - August 2019*
MSc., Mathematics, Advisor: Dr. Margaret Cheney GPA 3.54

- Master's thesis: Numerical Simulations for Synthetic Aperture Source Localization

University of St. Thomas St. Paul Minnesota *September 2012 - May 2017*
BA, Mathematics (Minors Mech. Eng. & Philosophy) GPA 3.76

TECHNICAL SKILLS

Mathematics	Modeling, numerical methods, estimation & detection, Monte Carlo methods, inference
Signal Processing	Distributed systems, RF modeling, HF/OTH radar, image processing, cyclostationary SP
Software	MATLAB, Python, Git, Linux, Parallel computing, Slurm, LaTeX, Mathematica, C++

SELECTED PUBLICATIONS

In Preparation (Preprints available upon request)

- **Liam J. Coulter**, Abhinav V. Sambasivan, Richard G. Paxman, and Jarvis D. Haupt, "Computational Information-Theoretic Analysis for Parameter Estimation in Optical Imaging Systems," *IEEE Transactions on Computational Imaging*, submitted May **2026**.
- Abhinav V. Sambasivan, **Liam J. Coulter**, Richard G. Paxman, and Jarvis D. Haupt, "On the Effects of Computational Forward Model Uncertainty in Parameter Estimation Lower Bounds," *IEEE Transactions on Information Theory* submitted June, **2026**.
- **Liam J. Coulter** and Matthew A. Bliss, "Resolving Ambiguity in Passive TDOA Geolocation Systems with an Altitude-Parametrized EKF," accepted for presentation at the **2026** 60th IEEE Asilomar Conference on Signals, Systems, and Computers.

Refereed Papers

- **Liam J. Coulter**, David J. Geroski and Andrew C. Marcum, "Joint Recursive Receiver and Emitter Geolocation Using Time Difference of Arrival Measurements", **2023** 57th IEEE Asilomar Conference on Signals, Systems, and Computers, Pacific Grove, CA, USA, 2023, pp. 1277-1282, doi: 10.1109/IEEECONF59524.2023.10476880.
- S. Gleason, I. Cherniak, I. Zakharenkova, D. Hunt, S. Sokolovskiy, D. Freesland, A. Krimchansky, J. McCorkel, **Liam Coulter**, G. Ramsey, and J. Chapel, "The First Atmospheric Radio Occultation Profiles From a GPS Receiver in Geostationary Orbit", IEEE Geoscience and Remote Sensing Letters, vol. 19, pp. 1-5, **2022**, Art no. 1005605, doi: 10.1109/LGRS.2022.3185828

Abstracts & Presentations

- **Liam Coulter**, J. McCorkel, and S. Gleason "Radio Occultation Measurements of Earth's Atmosphere from Geostationary Orbit for Weather Prediction and Water Vapor and Precipitation Monitoring," AMS Fifth Conference on Atmospheric Biogeosciences, June **2021**. Abstract published. Presentation received "Outstanding Oral Presentation" award.
- **Liam Coulter** and Margaret Cheney, "Synthetic Aperture Source Localization," SIAM Front Range Applied Mathematics (FRAM) graduate student conference presentation, March **2019**. Abstract published.